

Is the surface forcing in sea ice leads transferred to the Arctic ocean interior?

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Abstract

The Arctic sea-ice, in particular the ice-pack, acts as an insulator between the atmosphere and the ocean below. Leads, common in Arctic sea ice, facilitate exchanges of heat, salt, moisture and gases between the ocean and the atmosphere. Existing literature suggests that leads play a significant role in the heat budget of the surface of the ocean and likely impact water mass transformation within the Arctic. However, due to the lack of observations and high resolution models capable to resolve leads, an estimate of the impact of leads in the ocean buoyancy has been elusive. Here, we explore the magnitude and impact in the ocean surface caused by the leads using SEDNA, a subkilometer pan-Arctic hind-cast (1/60°; NEMO model). Leads account for 22% of the sea-ice cover extent, and within them, there is approximately 25% of the total surface water mass transformation. The buoyancy flux and water mass transformation in leads are similar to those underneath the surrounding ice covered oceans and the percentage of water mass transformation through leads matches their area coverage. Thus, the present estimate indicates that leads have a small contribution to Arctic Ocean dynamics, contrary to previous hypotheses.

Keywords: Arctic, Leads, Buoyancy flux

1 Introduction

The Arctic sea-ice ~~plays a crucial role in regulating~~ regulates Earth's climate by acting as a natural insulator between the atmosphere and the ocean ~~below~~. This insulating property acts as a thermal barrier, limiting heat loss from the ocean to the atmosphere and minimizing evaporation (Wettlaufer et al, 1997; Untersteiner, 1961). An ubiquitous feature of Arctic sea ice is the formation and persistence of leads. Leads occur across the polar oceans, both in the marginal ice zone (MIZ; 15 – 80% sea ice concentration) and the ice pack (> 80% sea ice concentration). ~~The difference between leads in these two regions arise from the mechanism driving their formation.~~ Within the MIZ, leads are primarily formed by advection of sea ice, while in the ice pack, leads form through deformation of the sea ice (~~ice pack leads are~~ commonly known in literature as “linear kinematic features”; LKFs). The main forcing of the sea ice advection and deformation ~~, thus the formation of leads,~~ is the wind (Linow and Dierking, 2017; Hutter et al, 2018), with a smaller contribution ~~by the~~ from ocean currents (Willmes et al, 2023). Leads can result in an opening ~~within of~~ the sea ice cover (Rampal et al, 2016; Hutchings et al, 2005; Richter-Menge et al, 2002) with spatial scales of ~~a few~~ meters to kilometers in width and a few kilometers up to hundreds of kilometers in length, and temporal scales ranging from a few hours up to a few days (Linow and Dierking, 2017; Wernecke and Kaleschke, 2015; Tschudi et al, 1998). Thus, they ~~are windows between the ocean and the atmosphere that can allow large flux exchanges, and can~~ locally impact the surface forcing ~~of in~~ the ice covered polar oceans (Lüpkes et al, 2008).

Once leads are formed, atmospheric forcing in conjunction with localized upwelling or downwelling occurring in the surface layer can result in important heat fluxes at the ocean surface ($\mathcal{O} \sim 100W/m^2$; Bourgault et al, 2020; McPhee et al, 2005; Marcq and Weiss, 2012; Maykut, 1986), instigating localized melting or freezing of sea ice (von Albedyll et al, 2022). For example, Boutin et al (2023) estimated that 35% of the total sea ice growth during winter occurs within the leads in the ice pack. The Arctic ice covered ocean is a β -ocean, i.e. the ocean stratification is mainly controlled by salinity ~~that,~~ which in turn is largely determined by ice-ocean interactions ~~when (sea-ice growths and melts)~~ growth and melt. Thus, leads experiencing an increase in buoyancy forcing due to brine rejection can induce convection (Smith and Morison, 1998), weaken the mixed layer stratification, and generate fronts, mixed layer turbulence, and eddies (Reiser et al, 2020; Smith et al, 2002). Meanwhile, leads experiencing melting will increase the mixed layer stratification and stabilize the ocean surface layer. These changes in the buoyancy flux translate into local transformation of the surface water masses that could be critical on the functioning of the Arctic Ocean circulation (Lenn et al, 2022; Pemberton et al, 2015; Smith et al, 1990). Previous studies have shown that instances of leads impact locally the ocean surface and the properties of the Arctic Ocean mixed layer. Nonetheless, the integrated impact leads have in the large scale buoyancy forcing at the ocean surface across the Arctic basin has not been quantified yet.

Here, we present the first estimate of the contribution of the buoyancy forcing and water mass transformation within leads in the Arctic Ocean, using the last year of a seven year long pan-Arctic hindcast run at a subkilometer resolution (SEDNA,

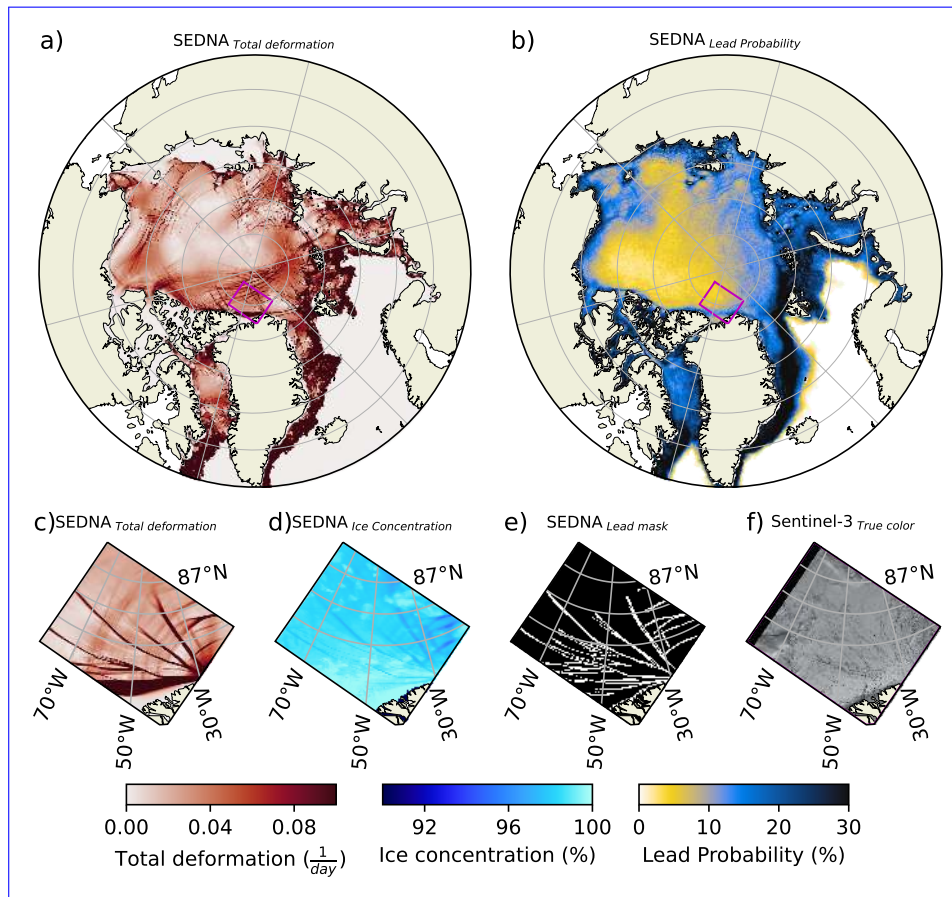


Fig. 1 a) Snapshot of total deformation for the 21st of April 2014 from SEDNA. b) Mean lead probability for daily snapshots of leads identified between 2011 and 2015 from SEDNA. c) True color image from Sentinel 3 on the 12th of April 2022 for the magenta area in panel a. d) Snapshot of the ice concentration from SEDNA on the 21st of April 2014. e) Total deformation to identify leads and f) identified mask of leads on the 21st of April 2014 in SEDNA.

Talandier and Lique, 2023; see Methods for ~~details of the calculations and numerical model~~ more details). Our main objectives are to: (1) assess the impact of leads in the surface buoyancy forcing and (2) contrast the surface water mass transformation within the leads and those of the ice cover excluding leads.

1 ~~Lead contribution to the buoyancy~~ Buoyancy forcing ~~at the ocean surface~~ in leads

Accurate representations of leads ~~require~~ are captured in models with horizontal resolutions ~~in the order of~~ a few kilometers (Bouchat et al, 2022; Wang et al, 2016);, however, to ~~adequately resolve the localized nature of~~ resolve localized freshwater and heat fluxes ~~across leads, higher horizontal resolutions in the subkilometer range~~ within

leads, subkilometer resolutions are needed. ~~To overcome this, here we analyse model~~
~~We address this by analyzing~~ outputs from the SEDNA ocean-sea ice model ~~SEDNA,~~
~~with a, which has an average Arctic~~ resolution of $\sim 800m$ ~~on average across the~~
~~Arctic capable,~~ sufficient to capture the ~~observational-observed~~ lead width distribu-
tion ($> 1km$; Wernecke and Kaleschke, 2015). A snapshot of the total deformation
is shown in figure 1a, where regions with locally high deformation values are identi-
fied as leads (See methods ~~for the definition of sea ice total deformation~~; Hutter et al,
2019; Kwok, 2001; Ruffieux et al, 1995). ~~Zooming into the magenta box~~ Upon closer
~~examination of the area~~ north of Greenland ~~;(panels c, d, and eof figure 4)~~, show
that regions of high deformation match those with lower ice concentration and the
identified lead mask. ~~Note that the sea ice concentration in the identified~~ Identified
leads shown in figure 1e using the lead identification algorithm proposed by Hutter
et al (2019) ~~remains retain ice concentrations~~ higher than 90% ~~concentration~~; this is a
consequence of continuous nature of the sea ice model and the daily averaged output
that smooths the presence of leads in the fields. Despite the high sea ice concentra-
tion, SEDNA ~~is capable to resolve intensified~~ captures localized sea ice deformation.
Furthermore, panels e and f reveal similarities between a typical spatial pattern of
observed leads on the 12th of April 2022 from a true color satellite image and the lead
mask from SEDNA for the 21st of April 2014.

The climatology of the lead probability, i.e. the likelihood of a leads occur-
ring for any given day, is shown in figure 1b. The mean probability of leads
averaged over the Arctic is $\sim 19\%$ in SEDNA, comparable to ~~observations using~~
~~RADARSAT Geophysical Processor System and Thermal-Infrared Satellite Imagery~~
~~previous observations~~ (Wernecke and Kaleschke, 2015; Willmes et al, 2023). Although
observations use different identification methods compared to the one implemented
in SEDNA (Hutter et al, 2019), there is resemblance between ~~the spatial patterns~~
~~of SEDNA and observations (Fig. S1)~~. We observe higher lead probability over the
shelves, along bathymetric features, and in regions with large kinetic energy (Fig. S1).
SEDNA reproduces the ~~patterns overall spatial patterns of lead probability~~ observed in
the Arctic ~~shelves, the Greenland Sea, Barents Sea, Kara Sea, Laptev Sea, and hotspots~~
~~in the Chukchi Sea, but the pattern in the Beaufort Sea in observations is not fully~~
~~captured in SEDNA, except for the Beaufort Sea~~. Regardless, the similarities between
the observations and the model lead probability give us confidence to quantify for the
first time the impact of leads in the surface buoyancy of the Arctic Ocean.

Figure 2a and b show two snapshots of the buoyancy flux at the ocean surface.
A negative buoyancy flux ~~in winter results mainly results~~ from sea ice refreezing and
brine rejection, while ~~in summer,~~ a positive buoyancy flux is associated with sea ice
melting and freshening of the Arctic surface. Inside the ice covered region (magenta
contour) in figure 2, the fluxes are spatially homogeneous, except for the marginal ice
zone and within the leads, where the buoyancy fluxes are larger. The area-weighted
buoyancy flux across the ice covered Arctic (15% - 100% ice concentration) within
the leads is $-2.2 \times 10^{-8} m^2/s^3$ and $7.2 \times 10^{-8} m^2/s^3$ for each date, compared to the
 $-2.4 \times 10^{-8} m^2/s^3$ and $5.7 \times 10^{-8} m^2/s^3$ excluding the leads. In other words, on the 1st
of January (winter), the total fluxes within the leads and those outside of the leads are
comparable. However, on the 25th of June (summer), the area weighted fluxes in the

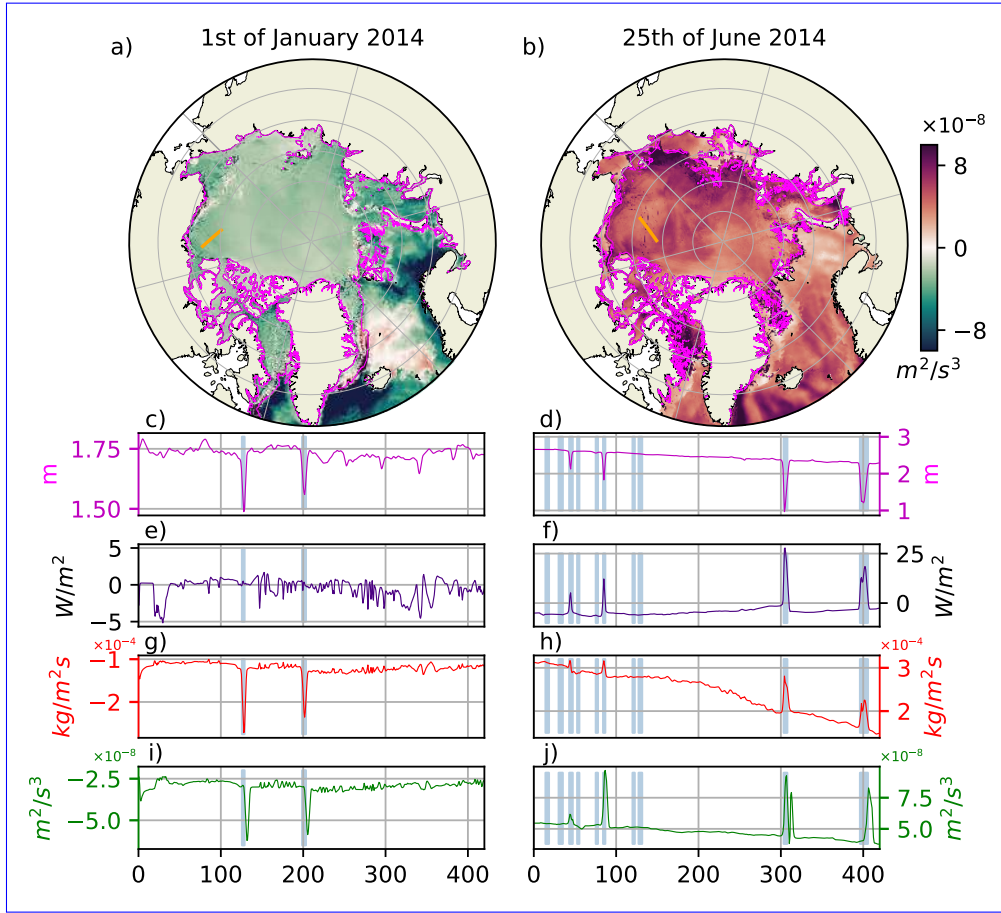


Fig. 2 Maps of buoyancy flux and transects for the 1st of January and 25th of June 2014. a) Map of buoyancy flux on the 1st of January 2014. Transect on the 1st of January 2014 of c) ice thickness, e) downward heat flux, g) minus freshwater flux, and i) buoyancy flux (heat flux minus freshwater flux). b) Map of buoyancy flux on the 25th of June 2014. Transect on the 25th of June 2014 of d) ice thickness, f) downward heat flux, h) minus freshwater flux, and j) buoyancy flux. Note that positive buoyancy fluxes correspond to buoyancy loss, while negative buoyancy fluxes correspond to buoyancy gain by the ocean. The transect location and orientation changes between the winter and summer dates. Magenta contour in panels a and b correspond to the 15% concentration contour of sea ice. Vertical blue bars shows the mask of leads in the transect.

leads can be up to 30% larger than those over the ice cover region ~~without considering~~ excluding leads.

~~Two transects across the Beaufort Gyre are shown in figure 2 for the 1st of January 2014 and 25th of June 2014. Examining the transect on the 1st of January 2014, we notice that along the transect we detect two leads (vertical blue bars). Within these detected leads there is~~ with a significant decrease in sea ice thickness of $\sim 25\text{cm}$, suggesting an opening of the sea ice cover (Fig.2 c). ~~Along this transect there is an overall~~ This transect exhibits small heat flux at the ocean surface and within

the identified leads (Fig.2 e), ~~since because~~ the ocean surface is at freezing point -
~~Thus, the extra and the~~ heat exchanged with the atmosphere is used to form sea
ice instead of further cooling the ocean, ~~resulting in a heat flux close to zero at the~~
~~ocean surface. In contrast, the.~~ The freshwater flux (Fig.2g) exhibits two prominent
positive peaks, located within the identified leads. Both ~~of these~~ peaks are twice the
magnitude of the fluxes ~~where no leads were identified, since there is outside the leads,~~
~~thus, highlighting~~ enhanced sea ice growth and brine rejection within the leads. The
buoyancy flux is negative for this snapshot where the largest buoyancy loss occurs
within the leads (Fig.2 i). On the 25th of June (Fig.2d), we detect several leads, but
only some show a decrease of the sea ice thickness of more than 1m (Fig.2 h). Leads
with a sea ice thickness decrease also experience an increase in heat fluxes, reaching
up to $\sim 25W/m^2$ (Fig.2f) and a local decrease of the freshwater flux, indicating a
freshening of the surface due to sea ice melt. This change is captured by the positive
buoyancy flux, decreasing the density at the surface (Fig.2j). ~~Note that some peaks~~
~~in the buoyancy flux align with the edge of the detected leads, potentially due to the~~
~~enhanced refreezing or melting that occurs at the edge of the lead or the advection~~
~~of the lead after it is formed.~~ Of all the identified leads, some exhibit significant
fluxes at the ocean surface, while others have little to no imprint in the thickness
and fluxes. ~~Detected leads without changes in the thickness or ocean surface fluxes~~
~~The latter ones~~ may be explained by the timing of the sea ice break up, where ~~later~~
~~subsequent~~ snapshots show an increase of fluxes at the location of these identified
leads (not shown). Alternatively, previous studies have shown that leads generated
through shear have a limited flux exchange with the atmosphere, but may still enhance
fluxes in the ocean ~~interior~~ due to isopycnal upwelling or downwelling forced by the
atmosphere-ocean stress within the leads (Bourgault et al, 2020).

Similar to observations, we identify fluxes that are larger within the leads than
outside of them for these summer and winter days. ~~The contribution of the freshwater~~
~~flux to the total buoyancy flux, attributable to melting and freezing, is significantly~~
~~larger than the heat flux contribution associated within the leads.~~ In the next section,
we explore the spatially averaged contribution of leads to the surface the buoyancy
fluxes and the integrated influence on the water mass transformation across the Arctic
Basin during 2014.

2 ~~Mean Integrated~~ contribution of ~~the~~ leads to the ~~surface~~ Arctic buoyancy

The ice covered Arctic experiences two main dynamical sea ice regimes, the marginal
ice zone (MIZ; 15-80% sea ice concentration) and ~~the~~ ice pack (80-100% sea ice concen-
tration). Thus, the following analysis are conducted separately for four distinct masks
representing regions with leads and regions excluding leads for the MIZ and the ice
pack. The averaged percentage area extent of these distinct masks during 2014 is as
follows: 6% for the MIZ, 72% for the ice pack, 6% for the leads in the MIZ, and 16%
for the leads in the ice pack. On average, the area extent covered by leads is $\sim 22\%$
of the total sea ice covered Arctic, consistent with previous model assessments (Wang
et al, 2016). The extent of each mask exhibits a seasonal cycle (Fig. 3a). From winter

(December, January, and February) to summer (June, July, and August), the percentage of the ice pack extent decreases from approximately 80% to 60%, while the extent of the MIZ increases from 2% to approximately 13%. On one hand, the percentage of the leads extent in the ice pack ice pack leads extent peaks in winter at almost 16% of the total surface covered by ice, with and decreases to a minimum extent of 13% in summer. On the other hand, the contribution of the leads in the MIZ covers up to 13% of the total ice cover Arctic in summer, and around 2% in winter. Overall the extend of leads in the MIZ is comparable to the MIZ extent excluding the leads extent of the MIZ leads covers 2% in winter and around 13% in summer, approximately the same as the MIZ extent.

The contribution of the total buoyancy flux occurring within the leads and ice covered regions ~~excluding leads~~ are quantified by weighting the buoyancy flux of each of the sea ice cover masks by the total ice covered area (Fig. 3b), ~~where;~~ note that the extent of each ~~ice cover~~ mask evolves in time. ~~When excluding the contribution within the leads, the~~. The mean buoyancy flux under the ice pack ~~excluding leads~~ in winter (December-February) and summer (June-August) is ~~$-1.8 \times 10^{-8} m^2/s^3$ and $2.2 \times 10^{-8} m^2/s^3$~~ $-1.8 \times 10^{-8} m^2/s^3$ and $2.2 \times 10^{-8} m^2/s^3$, respectively. The buoyancy flux underneath the ice pack changes sign during the seasonal cycle, suggesting a shift from ice formation in winter to ice melting in summer. Note that in the ice pack, the haline component of the buoyancy flux determines the buoyancy flux. Meanwhile, the mean buoyancy flux in the MIZ ~~(excluding the leads)~~ increases from a negligible ~~$0.02 \times 10^{-8} m^2/s^3$~~ $0.02 \times 10^{-8} m^2/s^3$ in winter to ~~$0.9 \times 10^{-8} m^2/s^3$~~ $0.9 \times 10^{-8} m^2/s^3$ in summer. The MIZ has predominantly a positive buoyancy flux suggesting preferential melting over the year (except for autumn), and ~~both thermal and haline component of the haline component also determines~~ the buoyancy flux ~~play a role in determining the buoyancy flux.~~ (See Fig.S2). In both the MIZ leads and the ice pack leads, the buoyancy flux follow the same seasonality ~~as the fluxes in the MIZ and the pack of the fluxes~~ outside of the leads. Winter values are ~~$-0.3 \times 10^{-8} m^2/s^3$ and $-0.01 \times 10^{-8} m^2/s^3$~~ $-0.3 \times 10^{-8} m^2/s^3$ and $-0.01 \times 10^{-8} m^2/s^3$, while summer values reach ~~$0.6 \times 10^{-8} m^2/s^3$ and $0.9 \times 10^{-8} m^2/s^3$~~ $0.6 \times 10^{-8} m^2/s^3$ and $0.9 \times 10^{-8} m^2/s^3$ for the contribution of the leads in the ice pack and the MIZ, respectively. ~~Comparing these fluxes weighted by the total ice covered area allows us to quantify their percentage contribution by region. In other words, buoyancy fluxes within the leads accounts for 15-30% of those underneath the ice pack, while in the MIZ leads, the buoyancy fluxes are comparable to those underneath the MIZ.~~ Overall, leads explain between 20% to 45percent-% of the ice covered buoyancy fluxes (red dashed line in Fig 3b). ~~This is despite the fact that the~~, more than the area extent of all leads that ranges from 18% in winter to 26% in summer ~~of the ice covered extend~~. Therefore, ~~it suggests that~~ fluxes are larger within the leads relative to ~~there~~ their area extent.

~~To further quantify this, the~~ The area weighted buoyancy fluxes by each region is shown in figure 3c. Buoyancy fluxes per unit of area exhibit two main different responses when comparing between the leads, and the MIZ and ice pack: i) winter leads in the MIZ and ii) summer leads in the ice pack. First, the mean winter (Fig. 3c) highlights the time periods in which fluxes are distinct within leads and those underneath the surrounding ice cover. During winter, the area weighted buoyancy flux

of the leads in the MIZ is MIZ leads is negative ($-0.8 \times 10^{-8} \text{m}^2/\text{s}^3$, a negative buoyancy flux in winter within the leads), compared to a positive in one outside the leads ($1.5 \times 10^{-8} \text{m}^2/\text{s}^3$). This suggests that within the leads there is. Thus, in winter leads are re-freezing ice, while underneath the ice-covered MIZ there is primarily melting. This is consistent with previous results that show more freezing occurs within leads in winter (Boutin et al, 2023). Second, the mean summer ice is melting. During summer, the area weighted buoyancy flux within the leads in the ice pack is $4.4 \times 10^{-8} \text{m}^2/\text{s}^3$ ice pack leads is $4.4 \times 10^{-8} \text{m}^2/\text{s}^3$, approximately 20% more buoyancy flux in summer compared to than the buoyancy flux underneath the ice pack ($3.6 \times 10^{-7} \text{m}^2/\text{s}^3$ per km^2 $3.6 \times 10^{-8} \text{m}^2/\text{s}^3$), indicating enhanced melting within the ice pack leads.

The role and impact leads have on the surface water masses of the Arctic can be quantified by using the surface water mass transformation framework (WMT; see methods section; Walin, 1982). This approach provides information on the transport of water through the surface density contours due to surface buoyancy forcing that results in the formation of lighter or denser water at the surface. Thus, the WMT relates the surface buoyancy forcing and the surface density field to the properties of the ocean interior. In figure 4a, the averaged WMT during 2014 is shown for the total ice covered Arctic, the ice pack, the MIZ, and the parts of these regions covered by leads. The definitions of the water masses depicted in figure 4a are taken loosely from those proposed by Pemberton et al 2015; Lansard et al 2012 Pemberton et al 2015 and Lansard et al 2012. The yearly mean WMT in the ice pack is characterized by a broad region of strong positive transformation (losing buoyancy) corresponding approximately to the halocline polar surface water (PSW_H ; $24.4 \text{kg}/\text{m}^3 < \rho < 27.4 \text{kg}/\text{m}^3$) reaching a maximum of $\sim 1.8 \text{Sv}$, consistent with previous estimates of the surface Arctic WMT (Pemberton et al, 2015). The yearly mean WMT in the ice pack leads also features a positive peak within the PSW_H reaching up to 0.4Sv . On average during 2014, the WMT in the ice pack leads is $\sim 20\%$ of the WMT in the ice pack (without leads). The mean MIZ WMT is mostly negative within the density range of $20 \text{kg}/\text{m}^3 < \rho < 27.5 \text{kg}/\text{m}^3$ (gaining buoyancy), reaching a minimum of -0.4Sv around $\rho = 27 \text{kg}/\text{m}^3$. The MIZ leads WMT is also negative within the same range of densities, with a minimum also of -0.4Sv . Thus, the MIZ leads WMT is comparable to those in the MIZ excluding leads. Maps of the surface density (Fig.S2) show that water mass transformation are localized geographically; the transformation of PSW_H occurs primarily within the Eurasian basin and a smaller contribution in the Chukchi sea, the PSW_{ML} ($\rho < 24.4 \text{kg}/\text{m}^3$) occurs in the Canadian basin, and the PDW and AW ($27.4 \text{kg}/\text{m}^3 < \rho < 27.8 \text{kg}/\text{m}^3$ and $27.8 \text{kg}/\text{m}^3 < \rho$, respectively) occurs in the Barents and Kara seas. The ice covered water mass transformation estimated from SEDNA, the addition of the WMT in the ice pack, the MIZ, and the leads, is consistent with previous estimates of the surface Arctic WMT (Pemberton et al, 2015).

Hovmöller diagrams of the seasonal cycle of the daily WMT estimates over 2014 for the ice pack, the MIZ, and leads in each of these regions are depicted in figure 4. The mean WMT underneath the ice pack varies from 1.1Sv in winter and -1.0Sv in summer, while the WMT in the leads in the ice pack is 0.2Sv in winter and -0.26Sv in summer. In other words, the WMT in the ice pack leads ranges from 20 to 25% of

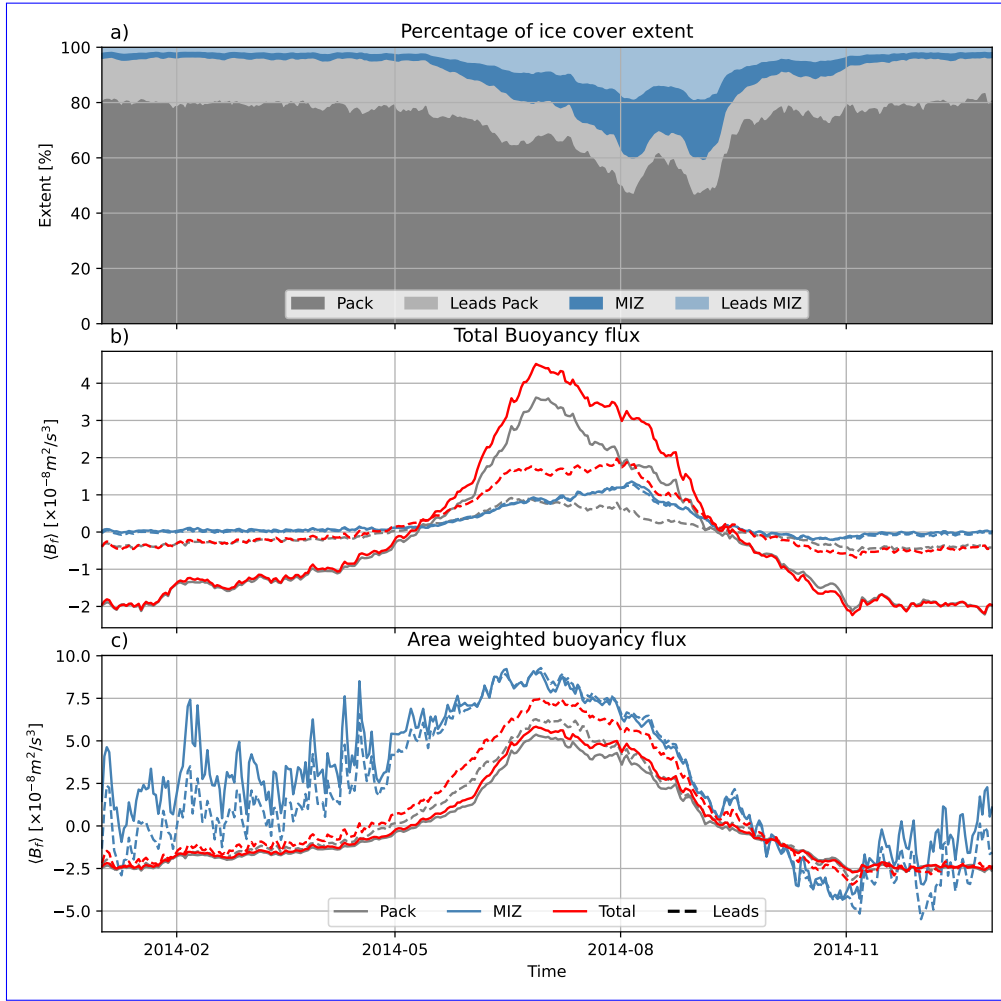


Fig. 3 Time-series during 2014 of a) the percentage of ice cover extent, and b) the area-weighted total buoyancy fluxes for the total ice covered, the ice pack, the MIZ, and the leads in the ice pack and in the MIZ, and c) the area-weighted buoyancy flux. The MIZ is defined between ice concentration of 15-80%, and the ice pack is defined where ice concentration > 80%, and the total ice covered is defined as the sum of the MIZ and ice pack.

the WMT occurring in the pack over the year (Fig. 4d). It is interesting to note that the patterns of the WMT across the density range for the leads and the surrounding regions excluding leads are similar (figures 4b and c). This suggests that there is not specific water mass that is formed transformed preferentially with the leads. Rather, our results suggests that the water masses formed transformed within the leads and outside of them have similar properties. In summer and winter, the The WMT in MIZ leads are comparable to and those in the MIZ WMT. In fact, have comparable magnitudes over the year the WMT are comparable in magnitude (Fig. 4 g). Similarly

to the leads in the ice pack, there is no evidence that leads in the MIZ could transform preferentially ~~some~~-specific water masses compared to the remaining MIZ. Adding the contribution of the WMT in all the leads in the Arctic (those in the MIZ and the ice pack), we estimate $\sim 25\%$ of the WMT ~~occurs in leads during the year. This happens in leads, and this~~ is comparable to the ~~mean~~-extent of leads in the Arctic ($\sim 22\%$). ~~The percentage of the WMT explained by leads vary seasonally. Around 10% to 25% of the full WMT occurs within leads during spring, autumn, and winter. However, in summertime the WMT in leads accounts for $\sim 45\%$ of the total WMT because the WMT in the MIZ dominates the signal and the MIZ leads transform the same volume of water as the MIZ excluding leads (See Fig. ??).~~

3 Conclusions

~~Leads are ubiquitous in the sea ice covered regions. They allow exchanges between the ocean and the atmosphere and locally enhance melting and freezing of sea ice.~~ The present study assesses the impact of leads in the ocean surface by diagnosing the buoyancy fluxes using a very high-resolution hindcast (SEDNA). The lead identification algorithm implemented in SEDNA allows us to reproduce the observed spatial distribution of leads over most regions of the Arctic, with exception of the Beaufort Sea, where the simulation underestimates the lead frequency. Likely because of the different identification algorithm, time periods, and spatial and temporal scales used in the present study and that of Willmes et al, 2023. Despite these regional differences, SEDNA can estimate the mean lead probability, showcasing its capacity to capture the dynamics of these features. ~~This high-resolution simulation allows us to better understand the role of leads in shaping the buoyancy fluxes of the Arctic Ocean.~~

Using SEDNA, we found that leads cover $\sim 22\%$ of the sea-ice covered Arctic, and in the ice pack, leads can contribute ~~up to 20%~~ more buoyancy flux compared to the fluxes outside the leads. Despite these larger buoyancy fluxes, leads only account for approximately $\sim 25\%$ of the total surface water mass transformation in the ice covered Arctic, which is roughly equivalent to their area coverage. Thus, contrary to previous hypotheses, our findings suggest that leads have a smaller contribution to the transformation of surface water masses in the Arctic Ocean than previously thought (e.g. Hutter et al, 2019; McPhee et al, 2005). While their presence can induce localized changes in the density of surface waters due to brine rejection and ice melting, ~~their frequency, spatial extent and duration are not sufficient to significantly alter the water masses. It has been argued that under sea ice cover in winter, brine rejection is not sufficient to penetrate the base of the mixed layer (Lenn et al, 2022), this also appears the case for leads that are unable of imprinting a signal in the Arctic Ocean interior. For example, we noted in SEDNA that the mixed layer depth (MLD) varies underneath the ice pack leads compared to regions outside the leads, however there is no evidence of an impact in the ice cover nor a response in the surface buoyancy or change of the water mass transformation within these regions. Furthermore, the transformation~~ the transformation of water masses in the leads have the same sign and magnitude as those underneath the surrounding ice-covered regions(~~excluding leads~~). Thus, no evidence was found that leads impact in a distinct way the Arctic ocean water masses in SEDNA.

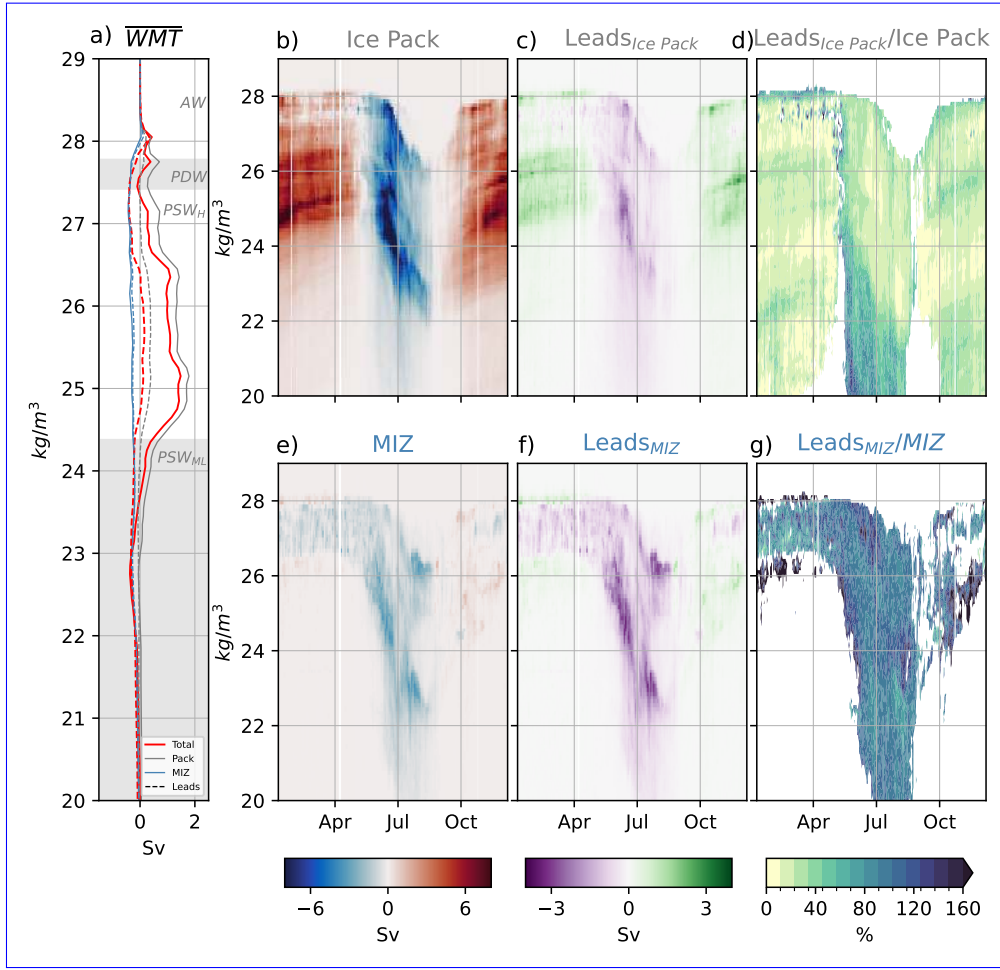


Fig. 4 Mean and daily Water mass transformation over 2014. a) Mean water mass transformation over 2014 for the total ice covered ice pack, MIZ, and leads in each region. Hovmöller diagram of the water mass transformation (Sv) for the b) ice pack, c) ice pack leads and d) the ratio of water mass transformation between the leads and the ice pack region. Panels e), f), and g) follow the same structure as panels b), c) and d), but for the MIZ, MIZ leads, and ratio between them. Shaded gray and white areas in panels a) represent approximately the water masses of the Arctic ocean according to [Pemberton et al 2015](#); [Lansard et al 2012](#): the mixed layer polar surface water ($PSW_{ML}; \rho < 24.4 \text{ kg/m}^3$), the halocline polar surface water ($PSW_H; 24.4 \text{ kg/m}^3 < \rho < 27.4 \text{ kg/m}^3$), the polar deep water ($PDW; 27.4 \text{ kg/m}^3 < \rho < 27.8 \text{ kg/m}^3$), and the Atlantic water ($AW; 27.8 \text{ kg/m}^3 < \rho$).

Here we focus on the surface forced WMT, but leads could further impact properties at the surface and interior of the ocean by enhancing mixing within leads, upwelling of isopycnals through Ekman pumping ([McPhee et al, 2005](#)), modifying entrainment in the mixed layer within the leads, providing a source of potential energy, and generating

mixed layer instabilities due to isopycnal tilting in the vicinity of leads. ~~A limitation of our study is the~~ The use of a continuous sea ice model that results in leads with high ice thickness and concentrations, ~~thus limiting limits~~ the exchanges between the ocean and the atmosphere. ~~In contrast~~ Therefore, future studies using floe resolving models that allow more “realistic leads” with direct exchanges between the ocean and the atmosphere would likely result in larger fluxes through the leads increasing their potential impact in the water mass transformation and ocean mixed layer.

Leads have an important local effect in the atmosphere and the sea-ice fluxes (Lüpkes et al, 2008; Marcq and Weiss, 2012; Wettlaufer et al, 1997; Smith et al, 1990), but a limited imprint on the ocean surface and interior properties. Our results suggest that the presence of leads is not the first order mechanism controlling the water mass transformation forced at the surface, thus leads have a minor impact in the stratification and surface layer dynamics of the Arctic Ocean. Therefore, resolving and/or parameterizing leads in climate models are likely to have a negligible impact improving the ~~well known issue within the Arctic climate modeling community of the~~ misrepresentation of the stratification and water masses in ~~the Arctic ocean~~ Arctic climate models (Wang et al, 2024; Ilicak et al, 2016). Finally, the probability of leads to occur in the ice pack has been projected to increase as the Arctic sea ice transitions towards thinner, younger, and more mobile sea ice (Boutin et al, 2023; Intergovernmental Panel on Climate Change (IPCC), 2022). Our analysis is only performed over one year and the future impact of more abundant leads is not represented in our simulation. Thus, dedicated studies are required to better estimate the surface buoyancy forcing within leads in the context of an evolving Arctic Ocean.

4 Methods

Model description

Here we employ the NEMO numerical platform (release 4.0.5; Madec et al, 2022) to establish a 1/60° pan-Arctic ocean-sea ice configuration named SEDNA. The ocean core (NEMO-OPA) solves the primitive equations using finite differences on an Arakawa C-grid mesh, and it is coupled to the SI3 sea-ice component (NEMO-SI3; NEMO Sea Ice Working Group, 2022). The pan-Arctic domain covers the Bering Strait on the Pacific side and extends southward to 56°N in the Subpolar North Atlantic and to the Baltic Sea entrance. The ocean model incorporates a linear free surface, utilizes a 3rd order flux form scheme for momentum advection, and employs a 4th order Flux Corrected Transport (FCT) for tracer advection. Additionally, the sea-ice representation uses an Elasto-Visco-Plastic (EVP) rheology with a 5-categories sea ice thickness distribution and a land-fast parameterization to better simulate sea-ice behavior above shelves.

The NCAR bulk formula from Large and Yeager (2009) is used to calculate air-sea fluxes based on the hourly ERA5 atmospheric data for near-surface variables (Hersbach et al, 2020). The three lateral boundaries are constrained with daily GLORYS12V1 Re-analysis data (Jean-Michel et al, 2021). Monthly freshwater fluxes from river discharges are combined with Greenland land ice melt data (Hu et al, 2019). The simulation extends over seven years (2009 to 2015) and starts from rest, utilizing initial conditions

based on the World Ocean Atlas 2009 temperature/salinity and mean January 2009 ice state from PIOMAS re-analysis (Locarnini et al, 2010; Zhang and Rothrock, 2003). The sea surface salinity is restored to the World Ocean Atlas climatology, but it is only applied ~~in~~ over the ice-free ocean. Preliminary assessment against available observations and re-analyses data shows that the simulation reasonably captures key ocean and sea-ice characteristics.

Identification of leads

The leads from SEDNA are identified by using the algorithm proposed by Hutter et al (2019). This method uses the total deformation rate defined as:

$$T_d = \left[\underbrace{\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \right)^2}_{Divergence} + \underbrace{\left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right)^2 + \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right)^2}_{Shear} \right]^{1/2}, \quad (1)$$

where u and v are the sea ice velocities. The deformation rate varies spatially depending on the background deformation and the sea ice properties (Reiser et al, 2020), where the largest deformation rates occur along the boundaries of ice floes where the leads are found. Therefore, to identify the leads, the local maxima is identified using a difference of Gaussian filter (DoG filter) of the deformation rate field. After this, a binary map is created with the mask of identified lead. Note that after this step, Hutter et al (2019) reduce the leads width to a line to then track the leads over time. Since our study focus on the ocean-atmosphere interactions, we only use the identification algorithm to produce a mask of leads during 2014 for the Arctic, thus we retain the original lead width computed from the deformation rate. Finally, the lead mask is then used to quantify and contrast the impact of leads compared to the ice covered Arctic, the ice-pack (ice concentration > 80%) and the MIZ (0 - 80% ice concentration).

Buoyancy flux and water mass transformation

Heat fluxes (shortwave radiation, longwave radiation, latent heat, and sensible heat) and freshwater fluxes (evaporation, precipitation, runoff, and sea ice melt/growth) at the ocean surface result in a buoyancy forcing capable to change the density of the ocean surface. Here, the buoyancy flux is computed as:

$$B_f = \underbrace{\frac{g\alpha Q_{net}}{\rho_0 C_p}}_{Net\ surface\ heat\ flux} - \underbrace{\frac{g\beta F_{net} SSS}{\rho_0}}_{Net\ surface\ freshwater\ flux}, \quad (2)$$

where Q_{net} the net heat flux at the sea surface, α the thermal expansion coefficient, SSS the sea surface salinity, F_{net} the net fresh water flux that includes evaporation, precipitation, runoff, and freezing minus melting flux, and β the haline contraction coefficient. With constant values of $g = 9.81 m/s^2$ and the ocean density $\rho_0 = 1026 kg/m^3$. Note that using this sign convention, positive values in the buoyancy

599 forcing decrease the surface density making the ocean surface more buoyant (stable)
600 and negative values increase the surface density. The area-weighted buoyancy ($\langle \rangle$)
601 fluxes are computed following:

$$602 \quad \langle B_f(t) \rangle = \frac{\sum_x \sum_y (B_f(t, x, y) * Area(x, y) * Mask_R(t, x, y))}{\sum_x \sum_y * Area(x, y) * Mask(t, x, y)}, \quad (3)$$

605 where the *Area* is be the area of the grid cells, and the *Mask* is the mask of MIZ, the
606 Pack, the MIZ leads, and the ice pack leads that vary in space and time. Note that
607 in Figure 3b *equation* (the divisor) corresponds to the full ice covered area (including
608 leads), while in Figure 3c *Mask* is equal to $Mask_R$.

609 Nurser et al (1999) and Walin (1982) proposed the water mass transformation
610 framework that quantifies the rate at which a given water mass is made lighter (nega-
611 tive transformation rate) or denser (positive) due to surface fluxes. The surface water
612 mass transformation is computed as:

$$613 \quad \Omega(\sigma_k) = -\frac{1}{\sigma_{k+1} - \sigma_k} \iint_A B_f dA, \quad (4)$$

614 where σ is potential density and the indices k represents the density bin number.
615 The computation was performed with 172 density bins of $0.1kg/m^3$ within the density
616 range of $15.15kg/m^3$ - $32.25kg/m^3$. The transformation rate Ω is spatially decomposed
617 into different sea-ice masks such as the ice covered, the pack ice, the MIZ, and the
618 leads in each of these regions.

621 **Open Source.** The SEDNA model configuration are described and publicly
622 available at <https://zenodo.org/records/7656924>. All analyses and figures in this
623 manuscript are reproducible via Jupyter notebooks and instructions can be found
624 in the GitHub repository Leads_Arctic_ocean (https://github.com/josuemtzmo/Leads_Arctic_ocean).
625

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633

634 635 636 References

- 637 von Albedyll L, Hendricks S, Grodofzig R, et al (2022) Thermodynamic and dynamic
638 contributions to seasonal Arctic sea ice thickness distributions from airborne obser-
639 vations. *Elementa: Science of the Anthropocene* 10(1). [https://doi.org/10.1525/](https://doi.org/10.1525/elementa.2021.00074)
640 [elementa.2021.00074](https://doi.org/10.1525/elementa.2021.00074)
641
642 Bouchat A, Hutter N, Chanut J, et al (2022) Sea Ice Rheology Experiment (SIREx):
643 1. Scaling and Statistical Properties of Sea-Ice Deformation Fields. *Journal of*
644 *Geophysical Research: Oceans* 127(4). <https://doi.org/10.1029/2021jc017667>

Bourgault P, Straub D, Duquette K, et al (2020) Vertical Heat Fluxes beneath Idealized Sea Ice Leads in Large-Eddy Simulations: Comparison with Observations from the SHEBA Experiment. *Journal of Physical Oceanography* 50(8):2189–2202. <https://doi.org/10.1175/jpo-d-19-0298.1>

Boutin G, Ólason E, Rampal P, et al (2023) Arctic sea ice mass balance in a new coupled ice-ocean model using a brittle rheology framework. *The Cryosphere* 17(2):617–638. <https://doi.org/10.5194/tc-17-617-2023>

Hersbach H, Bell B, Berrisford P, et al (2020) The era5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146(730):1999–2049. <https://doi.org/10.1002/qj.3803>

Hu X, Myers PG, Lu Y (2019) Pacific water pathway in the arctic ocean and beaufort gyre in two simulations with different horizontal resolutions. *Journal of Geophysical Research: Oceans* 124(8):6414–6432

Hutchings JK, Heil P, Hibler WD (2005) Modeling Linear Kinematic Features in Sea Ice. *Monthly Weather Review* 133(12):3481–3497. <https://doi.org/10.1175/mwr3045.1>

Hutter N, Losch M, Menemenlis D (2018) Scaling Properties of Arctic Sea Ice Deformation in a High-Resolution Viscous-Plastic Sea Ice Model and in Satellite Observations. *Journal of Geophysical Research: Oceans* 123(1):672–687. <https://doi.org/10.1002/2017jc013119>

Hutter N, Zampieri L, Losch M (2019) Leads and ridges in Arctic sea ice from RGPS data and a new tracking algorithm. *The Cryosphere* 13(2):627–645. <https://doi.org/10.5194/tc-13-627-2019>

Ilicak M, Drange H, Wang Q, et al (2016) An assessment of the arctic ocean in a suite of interannual core-ii simulations. part iii: Hydrography and fluxes. *Ocean Modelling* 100:141–161. <https://doi.org/10.1016/j.ocemod.2016.02.004>

Intergovernmental Panel on Climate Change (IPCC) (2022) *Polar Regions*, Cambridge University Press, p 203–320. <https://doi.org/10.1017/9781009157964.005>

Jean-Michel L, Eric G, Romain BB, et al (2021) The copernicus global 1/12° oceanic and sea ice glorys12 reanalysis. *Frontiers in Earth Science* 9. <https://doi.org/10.3389/feart.2021.698876>

Kwok R (2001) Deformation of the arctic ocean sea ice cover between november 1996 and april 1997: a qualitative survey. In: *IUTAM Symposium on Scaling Laws in Ice Mechanics and Ice Dynamics: Proceedings of the IUTAM Symposium Held in Fairbanks, Alaska, USA, 13–16 June 2000*, Springer, pp 315–322

691 Lansard B, Mucci A, Miller LA, et al (2012) Seasonal variability
 692 of water mass distribution in the southeastern beaufort sea deter-
 693 mined by total alkalinity and $\delta^{18}O$. *Journal of Geophysical Research:*
 694 *Oceans* 117(C3). <https://doi.org/https://doi.org/10.1029/2011JC007299>,
 695 <https://agupubs.onlinelibrary.wiley.com/doi/pdf/10.1029/2011JC007299>
 696
 697 Large W, Yeager S (2009) The global climatology of an interannually varying
 698 air-sea flux data set. *Climate Dynamics* 33(2-3):341–364. [https://doi.org/10.1007/](https://doi.org/10.1007/s00382-008-0441-3)
 699 [s00382-008-0441-3](https://doi.org/10.1007/s00382-008-0441-3)
 700
 701 Lenn YD, Fer I, Timmermans ML, et al (2022) Chapter 11 - mixing in the arctic ocean.
 702 In: Meredith M, Naveira Garabato A (eds) *Ocean Mixing*. Elsevier, p 275–299,
 703 <https://doi.org/https://doi.org/10.1016/B978-0-12-821512-8.00018-9>
 704
 705 Linow S, Dierking W (2017) Object-Based Detection of Linear Kinematic Features in
 706 Sea Ice. *Remote Sensing* 9(5):493. <https://doi.org/10.3390/rs9050493>
 707
 708 Locarnini R, Mishonov A, Antonov J, et al (2010) *World Ocean Atlas 2009, Vol. 1:*
 709 *Temperature*, vol 68, p 184
 710
 711 Lüpkes C, Vihma T, Birnbaum G, et al (2008) Influence of leads in sea ice on the
 712 temperature of the atmospheric boundary layer during polar night. *Geophysical*
 713 *Research Letters* 35(3). <https://doi.org/10.1029/2007gl032461>
 714
 715 Madec G, Bourdallé-Badie R, Chanut J, et al (2022) Nemo ocean engine. [https://doi.](https://doi.org/10.5281/zenodo.6334656)
 716 [org/10.5281/zenodo.6334656](https://doi.org/10.5281/zenodo.6334656)
 717
 718 Marcq S, Weiss J (2012) Influence of sea ice lead-width distribution on turbulent
 719 heat transfer between the ocean and the atmosphere. *The Cryosphere* 6(1):143–156.
 720 <https://doi.org/10.5194/tc-6-143-2012>
 721
 722 Maykut GA (1986) *The Surface Heat and Mass Balance*, Springer US, Boston, MA,
 723 pp 395–463. https://doi.org/10.1007/978-1-4899-5352-0_6
 724
 725 McPhee MG, Kwok R, Robins R, et al (2005) Upwelling of arctic pycnocline associated
 726 with shear motion of sea ice. *Geophysical Research Letters* 32(10). [https://doi.org/](https://doi.org/10.1029/2004GL021819)
 727 [10.1029/2004GL021819](https://doi.org/10.1029/2004GL021819)
 728
 729 NEMO Sea Ice Working Group (2022) Sea ice modelling integrated initiative (SI³) –
 730 the nemo sea ice engine. <https://doi.org/10.5281/zenodo.1471689>
 731
 732 Nurser AJG, Marsh R, Williams RG (1999) Diagnosing water mass formation from
 733 air-sea fluxes and surface mixing. *Journal of Physical Oceanography* 29(7):1468 –
 734 1487. [https://doi.org/10.1175/1520-0485\(1999\)029<1468:DWMFFA>2.0.CO;2](https://doi.org/10.1175/1520-0485(1999)029<1468:DWMFFA>2.0.CO;2)
 735
 736 Pemberton P, Nilsson J, Hieronymus M, et al (2015) Arctic Ocean Water Mass Trans-
 formation in S–T Coordinates. *Journal of Physical Oceanography* 45(4):1025–1050.

https://doi.org/10.1175/jpo-d-14-0197.1	737
Rampal P, Bouillon S, Ólason E, et al (2016) neXtSIM: a new Lagrangian sea ice model. <i>The Cryosphere</i> 10(3):1055–1073. https://doi.org/10.5194/tc-10-1055-2016	738
Reiser F, Willmes S, Heinemann G (2020) A New Algorithm for Daily Sea Ice Lead Identification in the Arctic and Antarctic Winter from Thermal-Infrared Satellite Imagery. <i>Remote Sensing</i> 12(12):1957. https://doi.org/10.3390/rs12121957	739
Richter-Menge JA, McNutt SL, Overland JE, et al (2002) Relating arctic pack ice stress and deformation under winter conditions. <i>Journal of Geophysical Research: Oceans</i> 107(C10):SHE 15–1–SHE 15–13. https://doi.org/10.1029/2000jc000477	741
Ruffieux D, Persson POG, Fairall CW, et al (1995) Ice pack and lead surface energy budgets during leadex 1992. <i>Journal of Geophysical Research: Oceans</i> 100(C3):4593–4612. https://doi.org/10.1029/94JC02485	742
Smith DC, Morison JH (1998) Nonhydrostatic haline convection under leads in sea ice. <i>Journal of Geophysical Research: Oceans</i> 103(C2):3233–3247. https://doi.org/10.1029/97jc02262	743
Smith DC, Lavelle JW, Fernando HJS (2002) Arctic Ocean mixed-layer eddy generation under leads in sea ice. <i>Journal of Geophysical Research: Oceans</i> 107(C8):17–1–17–17. https://doi.org/10.1029/2001jc000822	744
Smith SD, Muench RD, Pease CH (1990) Polynyas and leads: An overview of physical processes and environment. <i>Journal of Geophysical Research: Oceans</i> 95(C6):9461–9479. https://doi.org/10.1029/jc095ic06p09461	745
Talandier C, Lique C (2023) Sedna-delta. https://doi.org/10.5281/zenodo.7656924	746
Tschudi M, Curry J, Maslanik J (1998) Airborne observations of leads in the Beaufort Sea. <i>IGARSS '98 Sensing and Managing the Environment 1998 IEEE International Geoscience and Remote Sensing Symposium Proceedings (Cat No98CH36174)</i> 2:986–988 vol.2. https://doi.org/10.1109/igarss.1998.699648	747
Untersteiner N (1961) On the mass and heat budget of arctic sea ice. <i>Archiv für Meteorologie, Geophysik und Bioklimatologie, Serie A</i> 12(2):151–182. https://doi.org/10.1007/bf02247491	748
Walén G (1982) On the relation between sea-surface heat flow and thermal circulation in the ocean. <i>Tellus</i> 34(2):187–195. https://doi.org/10.1111/j.2153-3490.1982.tb01806.x	749
Wang Q, Danilov S, Jung T, et al (2016) Sea ice leads in the Arctic Ocean: Model assessment, interannual variability and trends. <i>Geophysical Research Letters</i> 43(13):7019–7027. https://doi.org/10.1002/2016gl068696	750

783 Wang Q, Shu Q, Bozec A, et al (2024) Impact of increased resolution on arctic ocean
784 simulations in ocean model intercomparison project phase 2 (omip-2). *Geoscientific*
785 *Model Development* 17(1):347–379. <https://doi.org/10.5194/gmd-17-347-2024>
786
787 Wernecke A, Kaleschke L (2015) Lead detection in Arctic sea ice from CryoSat-
788 2: quality assessment, lead area fraction and width distribution. *The Cryosphere*
789 9(5):1955–1968. <https://doi.org/10.5194/tc-9-1955-2015>
790
791 Wettlaufer JS, Worster MG, Huppert HE (1997) The phase evolution of Young Sea Ice.
792 *Geophysical Research Letters* 24(10):1251–1254. <https://doi.org/10.1029/97gl00877>
793
794 Willmes S, Heinemann G, Schnaase F (2023) Patterns of wintertime Arctic sea ice
795 leads and their relation to winds and ocean currents. *The Cryosphere Discussions*
796 2023:1–23. <https://doi.org/10.5194/tc-2023-22>
797
798 Zhang J, Rothrock DA (2003) Modeling global sea ice with a thickness and enthalpy
799 distribution model in generalized curvilinear coordinates. *Monthly Weather Review*
800 131(5):845 – 861. [https://doi.org/10.1175/1520-0493\(2003\)131<0845:MGSIIWA>2.0.](https://doi.org/10.1175/1520-0493(2003)131<0845:MGSIIWA>2.0.CO;2)
801 [CO;2](https://doi.org/10.1175/1520-0493(2003)131<0845:MGSIIWA>2.0.CO;2)
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./Figures/Figure_S1_comparison_LKF_observations_KE_bathy.pdf

Fig. S1 Comparison of winter lead frequency between a) Thermal-Infrared Satellite Imagery (Willmes et al, 2023) and b) SEDNA between 2011 and 2015. Panel c) shows the SEDNA mean kinetic energy between 2011 and 2015.

Seasonal average of the water mass transformation during 2014 for a) the ice covered, b) the ice pack, and c) the MIZ. Dashed lines correspond to the WMT for the leads in each of these regions.

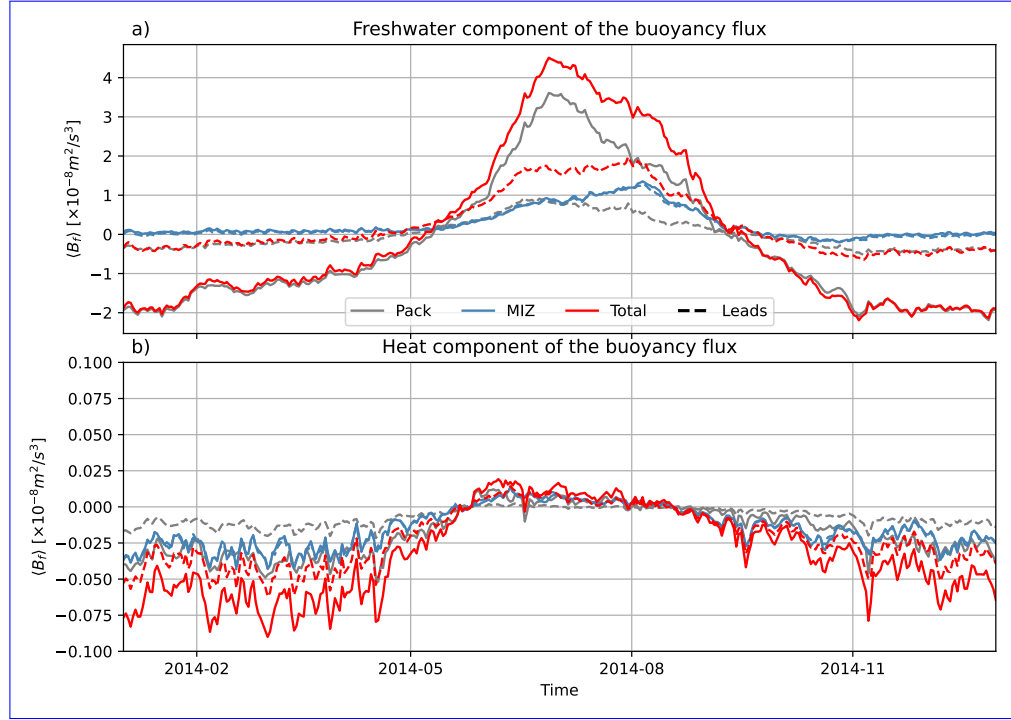


Fig. S2 Maps-Time-series during 2014 of a) the seasonal surface density-freshwater component of SEDNA during 2014. Solid the buoyancy flux and dashed magenta contour b) the heat component of the buoyancy flux for the ice pack, the MIZ, and the leads in maps correspond to the 15% ice pack and 80% in the MIZ. The MIZ is defined between ice concentration contour of sea 15-80%, and the ice respectively pack is defined where ice concentration > 80%.